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Technology program · Version 17.0 Revision 1

# Emerging Threats and Adaptive Protective Technology

Evidence first; rapid experimentation without unsafe deployment

## Threat-driven triggers for investment

Technology adoption starts only when a threat class is substantiated, an existing protective gap is documented and alternative human/process safeguards are compared. An emerging tactic may justify isolated research, not live autonomy. Revisit the threat register after relevant incidents, credible vendor findings, research or validated field failures.

| Candidate | Legitimate research question | Deployment boundary |
|---|---|---|
| AI-assisted triage | Can humans organize de-identified synthetic reports faster without more false escalation? | No prediction of an individual shooter or automatic police dispatch |
| Gunshot/acoustic detectors | Are false alarm and miss rates acceptable in site-appropriate, independently measured conditions? | No autonomous lockdown, suspect designation or force |
| Multimodal accessible alerts | Do students and staff receive consistent understandable protection messages under disruptions? | Accessible manual fallback and dispatch coordination required |
| Networked door and access systems | Do controls preserve code-compliant egress through outage or compromise? | Independent AHJ signoff before any live connection |
| Canine programs | For which specific screening task, if any, is performance validated and cost justified? | No general claim of shooting prevention; handler/welfare/rights compliance |
| Cyber defensive analytics | Can threats to critical systems be detected without student surveillance? | Constrain telemetry, vendor access and automatic changes |

## Procurement decision record

Record threat class, population exposure, baseline performance, prospective causal pathway, independent evidence, detailed failure effects, cybersecurity requirements, residual safety/rights risk, vendor lock-in and cost. A low-tech fix that performs as well with less harm should be preferred for that use case.

## Two-speed adoption

### Sandbox exploration

Synthetic data, physical isolation, opt-in non-identifying studies, rapid terminate/retrain decisions; no direct influence on live alerts or access control.

### Life-safety qualification

Independent evidence, cyber-physical validation, accessibility and fire review, human command authority, disaster recovery and renewal audits.

## Cybersecurity is a safety gate

Apply NIST CSF Govern, Identify, Protect, Detect, Respond and Recover. Require secure updates, strong identity controls, supplier accountability, reliable offline fallback and exercises for spoofed alerts, network loss and ransomware. Any remotely exploitable pathway to unsafe confinement blocks production approval.

## AI specific limitations

Dataset selection bias, synthetic deepfakes, hallucinations, domain shift and attacker adaptation must be explicitly tested. A model used for incident summaries cannot be assumed capable of predicting an attack or assigning probable guilt. Human reviewers own every consequential decision, and must be able to challenge model assertions.